

APEXPLANET SOFTWARE PVT. LTD. · DATA ANALYTICS INTERNSHIP

Data Storytelling & Statistical Validation

An end-to-end journey from raw sales data to a validated, data-driven business narrative



Final Capstone Presentation | Task 4 of 5

What We'll Cover Today

01



Dataset Overview

1,000 sales records across 12 attributes

02



Data Cleaning

Handling missing values & preparing the dataset

03



Exploratory Analysis

Uncovering patterns in category, city & gender

04



Interactive Dashboard

A live Power BI view of core KPIs

05



Hypothesis Testing

Does customer age drive sales differences?

06



Insights & Recommendations

What the business should do next

Dataset Overview

The analysis is built on ApexPlanet's 2025 sales transactions dataset, covering customer demographics, product details, and order value across every sale.

12 Fields Captured Per Order

- Order ID
- Order Date
- Customer ID / Name
- Age
- Gender
- City
- Product
- Category
- Quantity
- Unit Price
- Total Sales
- Discount

1,000

Total Orders

12

Data Fields

2025

Transaction Year

Preparing a Reliable Foundation

	Missing Age values	20 records	Filled with mean age (41.4 yrs)
	Missing City values	13 records	Labeled as "Unknown"
	Duplicate records	0 found	Dataset confirmed clean
	New field added	Discount	Flat 10% field engineered for analysis

KEY DESCRIPTIVE STATS

Age range
18 – 65 yrs (avg 41.4)

Quantity per order
1 – 10 units (avg 5.4)

Unit price
avg ₹25,487

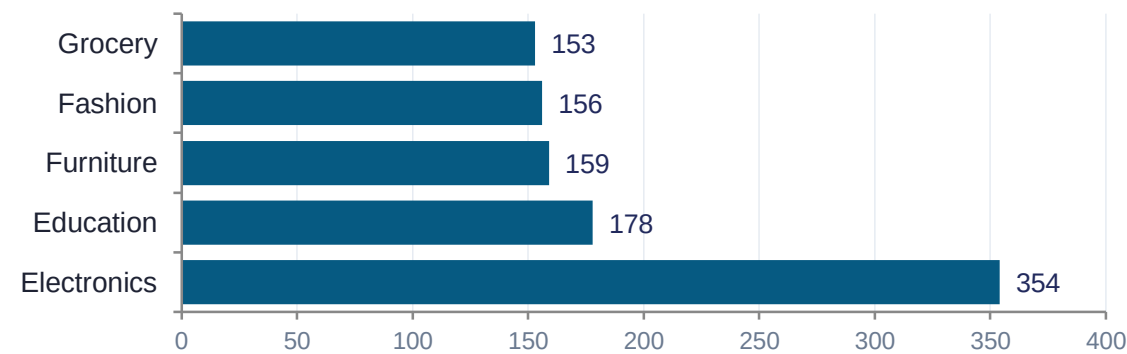
Total sales / order
₹437 – ₹4,93,678

Average order value
₹1,39,399

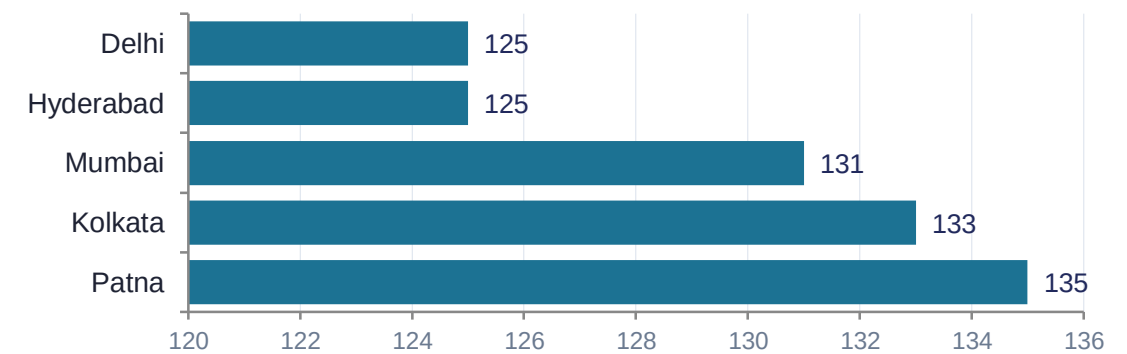
Exploring the Business Dimensions

The cleaned dataset (1,000 orders) was profiled across category, geography, product, and gender to see what shapes order volume.

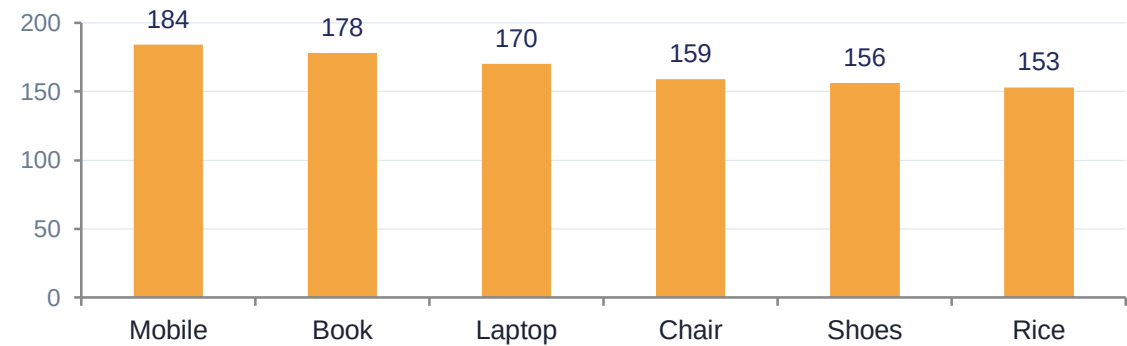
Orders by Category



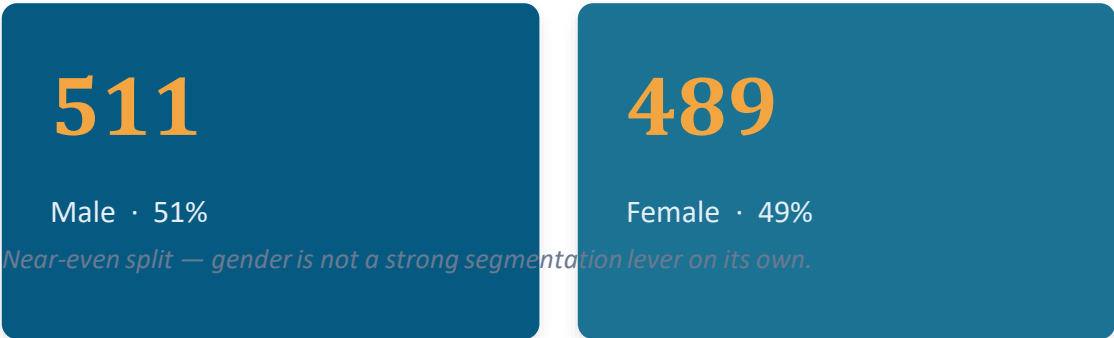
Top 5 Cities by Orders



Best-Selling Products



Gender Split



Customer Segmentation by Age Group

For the deep-dive, customers were segmented into five age bands to examine whether buying behavior varies across generations.



Customers per age group

CORE KPIS TRACKED



Total Sales

Overall revenue generated



Total Orders

Count of transactions processed



Average Order Value

Total Sales ÷ Total Orders



Category & City Mix

Explored via dashboard slicers

The Live Power BI Dashboard

A single, self-serve view of sales performance — fully filterable by city, category and gender.



Total Sales

KPI card



Total Orders

KPI card



Sales by Category

Bar chart



Sales by City

Column chart



Gender Distribution

Pie chart



Sales by Month

Line chart



Top 5 Cities

Ranked bar



Top 3 Categories

Ranked bar

Plus interactive slicers for City, Category, and Gender for on-the-fly filtering.

Hypothesis Testing: Does Age Drive Sales?



Hypothesis

Average total sales differ significantly across customer age groups.

STATISTICAL TEST USED

One-Way ANOVA (analysis of variance) across 5 age bands: 18–25, 26–35, 36–45, 46–55, 56+

Conclusion

With $p = 0.989$ ($\gg 0.05$), we fail to reject the null hypothesis. Customer age group does not have a statistically significant effect on total sales in this dataset.

0.078

F-Statistic

0.989

P-Value



Significance threshold $\alpha = 0.05$ — not met

Insights & Recommendations



A clean, reliable dataset

1,000 transactions with all missing values resolved and zero duplicates — a solid base for decision-making.



Self-serve visibility

The interactive dashboard lets stakeholders explore sales by category, city, and gender without needing a fresh analysis each time.



Age is not the differentiator

Statistical testing shows customer age group does not meaningfully explain differences in sales — segmentation efforts should look elsewhere.



Where to focus next

Prioritize deep-dives on category and city performance (from the dashboard's Top 5/Top 3 views) as more promising growth levers than age-based targeting.

Skills Demonstrated



Data Cleaning

Python & Pandas



Exploratory Analysis

Trend discovery



Dashboarding

Power BI



Statistical Testing

SciPy ANOVA



Data Storytelling

Stakeholder-ready decks

Thank you

Full project repository, source notebooks, and the live dashboard link are available in the master portfolio on GitHub.